Content-Author Guide: Setting Up a Clinical Trial in ZZedc

A complete role-based runbook for the study coordinator who provides the trial content

ZZedc Team

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Contents

About this guide

This is the canonical guide for the **non-technical team member** who provides the content of a clinical trial in ZZedc: the case-report forms, the validation rules, the user roster, the site assignments, and the day-to-day operating procedures. You do not need to write R code or run command-line tools. You will work alongside a **technical lead** (a statistician, biomedical informaticist, or IT-adjacent colleague) who handles installation, server configuration, and key management. That person follows a parallel guide, Technical-Lead Guide.

The guide is written for a first small-to-medium clinical trial (roughly 10 to 200 participants, 1 to 5 sites). It assumes you are setting up the trial from scratch. It does not assume any prior experience with ZZedc, R, or institutional IT.

By the end of this guide your trial will be ready for sites to begin collecting real subject data. The technical lead will have a working server; you will have a working data dictionary, validation rules, user roster, and trained site staff.

Chapter 1. Where you fit in the team

ZZedc separates the *trial content* (forms, rules, users) from the *technical infrastructure* (server, encryption, backups). This separation is intentional: it lets a domain expert author the content using tools they are comfortable with (Google Sheets), while a technical colleague handles the parts that require a terminal, database, or server.

Division of labour

You (content author)	Technical lead
Define the case-report forms	Install the package and provision the server
Author validation rules in plain language	Initialise the encrypted database
Build the user roster (names, sites, roles)	Run the import that loads your workbook
Test-drive the trial with synthetic data	Configure TLS, backups, and monitoring
Train site staff to use the application	Give site staff the URL and credentials
Resolve data queries with site coordinators	Maintain the system and verify the audit chain

You will exchange three artefacts with the technical lead: **(1)** the Google Sheets workbook URL, **(2)** the list of site-id values for your trial, **(3)** the agreed go-live date. Everything else lives in the workbook or in the application.

What if there is no technical lead?

For a single-investigator trial running locally on a laptop, the same person can play both roles. The Technical-Lead Guide walks through a 60-minute SQLite-on-laptop setup that is appropriate for solo work; if you can copy and paste a few R commands, you can follow it. For anything multi-site or multi-user, do not skip the technical lead.

Chapter 2. Trial-planning checklist before you start authoring

Spending an hour on this checklist saves a week of rework. Do not start filling in the workbook until you can answer every question.

Trial design

- ☐ What is the trial's full name and protocol identifier?
- ☐ How many participants will you enrol (a realistic ceiling for the next 12 months)?
- ☐ How many sites will collect data (1 for a single-site trial; otherwise list each)?
- ☐ What are the visit-points (Baseline, Week 4, Week 12, Month 6, …)?
- ☐ Which case-report forms are filled in at each visit (a Baseline-only form, a per-visit form, an at-discharge form)?
- ☐ Is the trial blinded, and if so, who sees what?

Forms and fields

For each case-report form on your list, sketch out:

- ☐ The form's short code (`enrollment`, `vitals`, `mmse`)
- ☐ The fields and their data types (text, number, date, single-choice, multi-choice)
- ☐ Which fields are required, which are optional
- ☐ Plausible ranges for each numeric field
- ☐ Allowed values for each single-choice field

You do not need to fill out the workbook yet. Sketch on paper or in a plain document.

Sites and roles

- ☐ List every site by short identifier (`SITE-001`, `SITE-002`, …) and full name
- ☐ For each site, list the people who will work on the trial and their roles (PI, Coordinator, Researcher, Monitor)
- ☐ Decide whether coordinators at one site should be able to see another site's data (usually no; default to siloed)

Validation rules

- ☐ Eligibility constraints (age range, exclusion criteria that can be expressed as a rule)
- ☐ Plausibility ranges for every numeric field (heart rate between 30 and 200, MMSE total between 0 and 30)
- ☐ Date orderings (date of birth before consent; randomisation before any visit)
- ☐ Visit-window rules (Week 4 visit must occur 25 to 31 days after randomisation)
- ☐ Cross-field consistency (if pregnant is `Yes`, sex must be `Female`; if a serious AE, a description is required)

These four lists are the input to the workbook. You can sketch them on paper, in a Word document, or in any tool you find convenient.

Chapter 3. Authoring the workbook in Google Sheets

This chapter is the core of your work. By the end of it you will have a Google Sheets workbook with three completed tabs (`Study_Users`, `Data_Dictionary`, `validation_rules`) ready to hand to the technical lead.

3.1 Before you begin

You will need:

- A Google account (work or personal).
- The workbook template URL from your technical lead. They will share a Sheet with you, or send you a template you can copy.
- Your sketches from Chapter 2.

You do **not** need:

- R, RStudio, or any command-line tool.
- A Google Cloud account.
- Any programming knowledge.

3.2 Create your workbook

If your technical lead sent you a template URL, click it and choose *File → Make a copy*. The copy lands in your own Google Drive. Rename it to something memorable, for example *MEMORY-001 Study Configuration*.

If you are starting from scratch, create a new Sheet at sheets.google.com and add three tabs. Right-click the tab strip at the bottom and choose *Rename*; name them `Study_Users`, `Data_Dictionary`, and `validation_rules` exactly as written. Capitalisation and spelling matter: the import tool looks for those tab names exactly.

3.3 Define your study team (`Study_Users` tab)

Click the `Study_Users` tab. The first row is for column headers; type these into row 1:

```
username | full_name | email | role | password_initial | site_id | active
```

Then start adding your team members, one per row.

Example: a four-person memory study at one site

username	full_name	email	role	password_initial	site_id	active
dr_smith	Jane Smith	jsmith@example.org	PI	TempPass!2026	001	TRUE
coord_jones	Bob Jones	bjones@example.org	Coordinator	TempPass!2026	001	TRUE
nurse_lee	Carol Lee	clee@example.org	Coordinator	TempPass!2026	001	TRUE
analyst_park	Diego Park	dpark@example.org	Researcher	TempPass!2026	001	TRUE

What each column means

- **username**: login name, no spaces, all lowercase. Once set, it should not change.
- **full_name**: how the person should appear in the audit log and reports.
- **email**: used for password resets and notifications.

- **role**: choose one of `PI`, `Coordinator`, `Researcher`, `DataManager`, `Monitor`, `Admin`. The role controls what the user is allowed to do. If you are unsure, `Coordinator` is the most common choice for nurses and study coordinators.
- **password_initial**: a starting password. The user is forced to change it the first time they log in. Use the same value for everyone.
- **site_id**: the site the user belongs to (see §3.4 for multi-site).
- **active**: `TRUE` for current team members, `FALSE` to disable an account without deleting the row.

Role cheat sheet

Role	Can do	Cannot do
`Admin`	Everything, including user management	(no restrictions)
`PI`	Read all subjects at their site, sign-off on data	User management
`Coordinator`	Enter and edit subject data at their site	See other sites
`Researcher`	Read and export data	Edit subject data
`Monitor`	Read-only audit and quality reports	Edit anything
`DataManager`	Edit, query, lock data; manage validation rules	Run the server

3.4 Multi-site authoring

For a multi-site trial, the `site_id` column is the only thing that changes. Each user belongs to exactly one site (or to "all sites" via the special `site_id = "ALL"`, reserved for the PI, central monitor, and data manager).

Example: a three-site memory trial

username	full_name	email	role	password_initial	site_id	active
dr_smith	Jane Smith	jsmith@example.org	PI	TempPass!2026	ALL	TRUE
ucsd_coord	Anh Nguyen	aN@ucsd.edu	Coordinator	TempPass!2026	UCSD	TRUE
ucsd_pi	Maria Lopez	ml@ucsd.edu	PI	TempPass!2026	UCSD	TRUE
jhu_coord	Sam Patel	sp@jhu.edu	Coordinator	TempPass!2026	JHU	TRUE
stan_coord	Tara Liu	tl@stanford.edu	Coordinator	TempPass!2026	STAN	TRUE
central_dm	Kim Tran	kt@central.org	DataManager	TempPass!2026	ALL	TRUE
central_mon	Pat Diaz	pd@central.org	Monitor	TempPass!2026	ALL	TRUE

Site-id naming convention

Use short, uppercase, alphabetic codes for site identifiers (`UCSD`, `JHU`, `STAN`, `BWH`). Numeric codes (`001`, `002`) work too but are less self-documenting. Whatever you choose, agree on the list with the technical lead before authoring, and reuse the exact same codes throughout the workbook.

Site-roster sub-tab (recommended)

Create an additional sub-tab named `Sites` with the columns `site_id`, `site_name`, `pi_name`, `pi_email`, `irb_number`, `active`. The technical lead can read this for reporting; the import does not require it but the document is essential for day-2 operations.

3.5 Define your forms (`Data_Dictionary` tab)

This is the longest of the three tabs. Each row describes one field on one form. If your study has three forms with ten fields each, this tab has thirty rows.

Click the `Data_Dictionary` tab and add these column headers in row 1:

```
form_code | field_name | field_label | field_type | required | validation | description | field_order
```

What each column means

- **form_code**: short name for the form, in lowercase, no spaces. Examples: `enrollment`, `vitals`, `mmse`.
- **field_name**: short, lowercase variable name for the field. No spaces. Examples: `subject_id`, `dob`, `mmse_total`.
- **field_label**: the human-readable label that appears next to the input box on screen.
- **field_type**: one of `text`, `numeric`, `date`, `select`, `radio`, `checkbox`. See the cheat sheet below.
- **required**: `TRUE` if the field must be filled in, `FALSE` otherwise.
- **validation**: optional. A short rule like `between 0 and 30` or `in(Yes, No)`. We expand on these in §3.6.
- **description**: help text that appears under the field for the data-entry user.
- **field_order**: a number from 1 upward, controlling the order in which fields appear. Use 10, 20, 30 to leave room for inserting fields later.

Field-type cheat sheet

Use this type	When the answer is…
`text`	A short free-text answer (name, ID, comment)
`numeric`	A number (age, weight, score)
`date`	A calendar date
`select`	One choice from a long drop-down list
`radio`	One choice from 2-5 visible options
`checkbox`	A yes/no answer or a multi-select list

Example A: Enrollment form

form_code	field_name	field_label	field_type	required	validation	description	field_order
enrollment	subject_id	Subject ID	text	TRUE		Format: SUBJ-NNN	10
enrollment	enrollment_date	Enrollment Date	date	TRUE		Date the consent was signed	20
enrollment	dob	Date of Birth	date	TRUE			30
enrollment	sex	Sex	radio	TRUE	in(Female, Male, Other)		40

form_code	field_name	field_label	field_type	required	validation	description	field_order
enrollment	education_years	Years of Education	numeric	TRUE	between 0 and 30	Total years of formal education	50
enrollment	medical_history	Notable Medical History	text	FALSE		Free text, brief	60

Example B: Vital signs form

form_code	field_name	field_label	field_type	required	validation	description	field_order
vitals	visit	Visit	select	TRUE	in(Baseline, Week 4, Week 8)		10
vitals	exam_date	Examination Date	date	TRUE			20
vitals	weight_kg	Weight (kg)	numeric	TRUE	between 30 and 200		30
vitals	sbp	Systolic BP (mmHg)	numeric	TRUE	between 70 and 220	Average of two readings	40
vitals	dbp	Diastolic BP (mmHg)	numeric	TRUE	between 40 and 130		50
vitals	hr	Heart Rate (bpm)	numeric	TRUE	between 30 and 200		60

Example C: MMSE assessment

form_code	field_name	field_label	field_type	required	validation	description	field_order
mmse	visit	Visit	select	TRUE	in(Baseline, Week 4, Week 8)		10
mmse	assessment_date	Assessment Date	date	TRUE			20
mmse	orientation_time	Orientation to Time	numeric	TRUE	between 0 and 5	Score 0-5	30
mmse	orientation_place	Orientation to Place	numeric	TRUE	between 0 and 5	Score 0-5	40
mmse	registration	Registration	numeric	TRUE	between 0 and 3	Score 0-3	50
mmse	attention	Attention and Calculation	numeric	TRUE	between 0 and 5	Score 0-5	60
mmse	recall	Recall	numeric	TRUE	between 0 and 3	Score 0-3	70
mmse	language	Language	numeric	TRUE	between 0 and 8	Score 0-8	80

form_code	field_name	field_label	field_type	required	validation	description	field_order
mmse	visual_spatial	Visual-Spatial	numeric	TRUE	between 0 and 1	Score 0-1	90
mmse	mmse_total	MMSE Total	numeric	TRUE	between 0 and 30		100

Example D: Adverse event log

form_code	field_name	field_label	field_type	required	validation	description	field_order
ae	ae_term	Adverse Event	text	TRUE		Brief description	10
ae	onset_date	Onset Date	date	TRUE			20
ae	severity	Severity	radio	TRUE	in(Mild, Moderate, Severe)		30
ae	relationship	Study-Drug Relation	radio	TRUE	in(Unrelated, Unlikely, Possible, Probable, Definite)		40
ae	serious	Meets SAE Criteria?	checkbox	TRUE			50
ae	resolution_date	Resolution Date	date	FALSE		Leave blank if ongoing	60

3.6 Define validation rules (`validation_rules` tab)

Validation rules catch data-entry errors before they propagate. A rule is a short condition that must be true for the data to be saved. If the condition is false, ZZedc shows an error and asks the user to correct the entry.

Click the `validation_rules` tab and add these column headers in row 1:

```
rule_id | field_code | rule_dsl | form_code | rule_name | error_message | severity | rule_category | is_
```

What each column means

- **rule_id**: unique short label, in upper case. Examples: `AGE_RANGE`, `MMSE_SUM`, `DOB_BEFORE_CONSENT`.
- **field_code**: the field the rule applies to. Match exactly what you put in the `field_name` column on the `Data_Dictionary` tab.
- **rule_dsl**: the rule itself, in plain English (see below).
- **form_code**: which form the rule belongs to.
- **rule_name**: a human-readable name shown in reports.
- **error_message**: what the data-entry user sees when the rule fails. Be specific and constructive.
- **severity**: `ERROR` (blocks save), `WARNING` (saves but flags), `INFO` (shows a note). When in doubt, use `ERROR`.
- **rule_category**: `FIELD` for single-field checks, `CROSS_FIELD` for checks involving multiple fields on the same form, `CROSS_FORM` for checks across forms.
- **is_active**: `TRUE` for rules that should fire, `FALSE` to temporarily disable a rule without deleting it.

Six validation patterns to copy

1. Range check The field must fall in a numeric range.

```
age between 18 and 90
mmse_total between 0 and 30
weight_kg between 30 and 200
```

2. List membership The field must be one of a small set of allowed values.

```
sex in (Female, Male, Other)
severity in (Mild, Moderate, Severe)
visit in (Baseline, Week 4, Week 8)
```

3. Date ordering

```
dob < enrollment_date
enrollment_date <= randomization_date
randomization_date <= assessment_date
resolution_date >= onset_date
```

4. Visit window

```
assessment_date - randomization_date between 25 and 31
```

(Week 4 visit must occur between 25 and 31 days after randomisation.)

5. Conditional consistency

```
if pregnant == "Yes" then sex == "Female" endif
if serious == TRUE then sae_criteria != "" endif
if visit == "Baseline" then visit_number == 1 endif
```

6. Sum / formula

```
mmse_total = orientation_time + orientation_place + registration + attention + recall + language + visua
```

A complete example

rule_id	field_code	rule_dsl	form_code	rule_name	error_message	severity	rule_category	is_active
AGE_RANGE	dob	(enrollment_date - dob) / 365 between 50 and 90	enrollment	Age at enrollment 50-90	Subject must be aged 50 to 90 at enrollment	ERROR	CROSS_FIELD	TRUE
EDU_RANGE	education_years	education_years between 0 and 30	enrollment	Education 0-30 years	Education must be 0-30 years	ERROR	FIELD	TRUE
BP_RANGE	sbp	sbp between 70 and 220	vitals	Systolic BP plausible	Systolic BP outside 70-220 mmHg	ERROR	FIELD	TRUE

rule_id	field_code	rule_dsl	form_code	rule_name	error_message	severity	rule_category	is_active
BP_ORDER	sbp	sbp > dbp	vitals	Systolic > Diastolic	Systolic must be greater than diastolic	ERROR	CROSS_FIELD	TRUE
MMSE_TOTAL	mmse_total	mmse_total between 0 and 30	mmse	MMSE total in range	MMSE total must be 0-30	ERROR	FIELD	TRUE
MMSE_SUM	mmse_total	mmse_total = orientation_time + orientation_place + registration + attention + recall + language + visual_spatial	mmse	MMSE component sum	MMSE total does not match component sum	ERROR	CROSS_FIELD	TRUE
WK4_WINDOW	assessment_date	assessment_date - randomization_date between 25 and 31	mmse	Week 4 visit window	Week 4 visit must occur 25-31 days after randomization	WARNING	CROSS_FORM	TRUE
AE_RESOLVED	resolution_date	if resolution_date != “ ” then resolution_date >= onset_date endif	ae	Resolution after onset	Resolution date must be on or after onset date	ERROR	CROSS_FIELD	TRUE
SAE_REQUIRES_DESC	ae_description	if serious == TRUE then ae_description != “ ” endif	ae	SAE requires description	Serious adverse events must include a description	ERROR	CROSS_FIELD	TRUE

You do not need every rule on day one. Start with range checks and required fields, and add cross-field rules as the study matures.

3.7 Share the workbook with your technical lead

Once the three tabs are filled in, click *Share* in the upper right of Google Sheets and share the workbook with your technical lead's Google account. Editor permission is appropriate.

Send the technical lead:

- The URL of the workbook (copy it from the address bar).
- The intended database file name, if you have one in mind.
- Whether this is a fresh setup or an update to an existing database.

The technical lead will run a single R command, `setup_zzedc_from_gsheets()`, that reads your three tabs and populates the ZZedc database. Expected time: under one minute for a small study.

Chapter 4. Test-driving the trial

Before site staff touch the application, *you* should walk through one synthetic subject end-to-end. This is the single highest-value quality check you can do. It catches every problem that paper review will miss: a misspelled field name, a validation rule that was supposed to be a warning but got set to error, a confusing label, a missing required field.

The dry-run protocol

After the technical lead confirms the import succeeded, ask them for the application URL and your `Admin` credentials. Then:

1. **Log in as Admin.** Confirm you see all forms in the sidebar.
2. **Create a synthetic subject.** Use a fake ID (`TEST-001`). Do not use any real patient data.
3. **Walk through every form, every field, every visit.** Enter realistic but synthetic values.
4. **Test the "wrong" path.** Try to enter a value that should trigger each validation rule. Confirm the error message is helpful.
5. **Save and re-open the subject.** Confirm the data persists.
6. **Run a basic export.** Choose CSV; open it in a spreadsheet; confirm every field appears with the value you entered.
7. **Log in as a Coordinator at the wrong site.** Confirm you cannot see your synthetic subject.
8. **Delete the synthetic subject.** Confirm it disappears from the list (a record of the deletion remains in the audit log; this is intentional).

If anything looks wrong, do not paper over it. Tell the technical lead, and either re-edit the workbook (for form/rule/user changes) or have the technical lead fix the configuration directly.

What "looks wrong" usually means

Symptom	Likely cause
A field is missing from a form	Wrong `form_code` or missing row in `Data_Dictionary`
A validation rule never fires	`field_code` typo or `is_active` set to `FALSE`
Site siloing is wrong	Wrong `site_id` on a user or on the rule
A drop-down is missing options	`validation` cell uses wrong syntax (`in(A, B, C)` is correct)
The audit log shows no entries	Technical-lead issue, not a content issue

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Chapter 5. Training site staff

Train each site separately

Plan a 30 to 45 minute training session for each site. Schedule it within a week of go-live, not earlier (people forget). Do not train multiple sites at once unless they share staff.

Training agenda (30 minutes)

1. **Trial recap (5 min).** What we are studying, what data we collect at each visit, what counts as an adverse event.
2. **Live demo (10 min).** Open the application; log in; create a synthetic subject; fill in the Baseline form; trigger one validation error on purpose; save; re-open.
3. **Each coordinator does one synthetic subject (10 min).** They drive; you watch over their shoulder. Use a different fake ID per coordinator (`TEST-SITE-001`, `TEST-SITE-002`).
4. **Questions and red flags (5 min).** Anything that confused anyone is a sign that the form, label, or description text needs revision.

What to hand over

- The application URL.
- Each coordinator's username (you do not see their password; they will set it on first login).
- A one-page printed cheat sheet with: how to log in, how to create a subject, how to enter a visit, how to ask for help.
- Your phone or email for the first two weeks (so problems do not pile up while site staff figure out who to ask).

Common training gotchas

- **First login.** Coordinators are forced to change the initial password. Some get nervous and call you. Reassure them: a strong password they choose is the right answer.
- **Browser compatibility.** ZZedc works in any modern browser. Internet Explorer 11 is not modern; ask sites running it to use Edge or Chrome.
- **Required fields.** People sometimes assume optional fields are required because the form is long. Print the cheat sheet with the required-vs-optional distinction visible.
- **Form layout on mobile.** The application is responsive but not optimised for phones. Site coordinators should use a laptop or tablet.

Chapter 6. Day-to-day operations

Once sites begin entering subjects, your job shifts from *authoring* to *operating*. This chapter covers the procedures that keep the trial running smoothly.

6.1 Reviewing data weekly

Block 30 to 60 minutes per week to review newly entered data. Open the application, switch to the *Data Explorer* tab, and look at:

- New subjects since last week.
- Visits completed since last week.
- Validation warnings (rules with `severity = WARNING` that fired but did not block save).
- Any subjects with no follow-up visit recorded but a baseline more than (visit-window-end + 7 days) ago.

The technical lead can configure an automated weekly summary e-mail; ask for it if you do not have one.

6.2 Resolving data queries

A *query* is a question you raise about a specific data point. Common reasons: a value is implausible (a 27-year-old in a 50-90 trial), a date is missing, a follow-up visit is overdue.

The query life cycle

1. **Open** the query in the application: navigate to the subject, click the field in question, choose *Raise query*. Write a specific question (*"Is the height value 1.65 in metres or 165 in cm?"*).
2. The site coordinator gets the query in their *Open Queries* list at next login.
3. **Respond.** The coordinator either corrects the value (with a reason) or explains why the value is correct.
4. **Close** the query. You read the response, accept it (or send it back if more clarification is needed), and close.

Query SLA

Agree on a service-level expectation with the site coordinators at training. Suggested:

Query type	Response expected	Closure expected
Plausibility check	3 business days	5 business days
Missing required value	1 business day	3 business days
SAE-related	24 hours	24 hours

Record the SLA in the protocol and on the cheat sheet.

6.3 Locking completed visits

Once a subject completes a visit and all queries are resolved, the visit can be *locked*. Locked data cannot be edited without a documented data-correction request, which the audit log records. Use the *Lock Visit* control in the application.

Lock visits as soon as they are clean. This protects the trial from late-stage edits that compromise data integrity.

6.4 Exporting data for analysis

When the analyst needs a snapshot for an interim analysis or for the final report:

1. Tell the technical lead you need an export.
2. They will produce a snapshot in CSV, Excel, or SAS XPT format from the *Export* tab and place it in the agreed secure-share location.
3. The audit log records the export, who ran it, and what was in it. The analyst inherits the obligation to handle the data per the protocol.

Do not export the database directly; always go through the application's *Export* tab so the action is logged.

Chapter 7. Pre-go-live checklist

Use this checklist 7 to 14 days before the planned go-live date.

Content readiness

- ☐ All case-report forms are in the `Data_Dictionary` tab, with required-vs-optional set correctly
- ☐ All validation rules are in the `validation_rules` tab and active
- ☐ Every site is represented in `Study_Users` with at least one PI and one Coordinator
- ☐ The roster has been reviewed by the trial PI and approved
- ☐ You have walked through one synthetic subject end-to-end (Chapter 4)

Technical readiness (verify with the technical lead)

- ☐ Server is provisioned and reachable from each site
- ☐ TLS / HTTPS is enabled (the URL starts with `https://`)
- ☐ Daily backups are configured and tested (the technical lead has performed at least one restore-from-backup drill)
- ☐ Encryption at rest is enabled and the encryption key is held by the agreed custodian
- ☐ The audit log validates end-to-end (the technical lead has run `verify_audit_log_integrity()`)

People readiness

- ☐ Each site has had its 30-minute training session
- ☐ Each coordinator has logged in once with their initial credentials and changed their password
- ☐ Each coordinator has created their own synthetic subject successfully
- ☐ You have a documented escalation path (coordinator → you → technical lead → PI)
- ☐ The query SLA is agreed and documented

Trial-specific readiness

- ☐ The IRB has approved the final CRF set
- ☐ The protocol references the correct ZZedc CRF version
- ☐ Sponsor (or DCC, or PI) has signed off on go-live

When every box is checked, schedule a 30-minute go-live call with the technical lead and the site PIs. On the call: confirm the URL, confirm the credentials work, ask each site PI for "go" or "no-go", and switch the application to the *Live* banner state via the technical lead's command.

Chapter 8. After go-live

The first two weeks after go-live are the highest-risk window. Plan for it.

Week 1

- Daily check-in with each site coordinator (5 minutes by phone or message).
- Daily review of new subjects in the application.
- Same-day query resolution for anything that comes up.

Weeks 2 to 4

- Weekly check-in with each coordinator.
- Weekly review of new data.
- 24- to 48-hour query-resolution turnaround.

Steady state (week 5 onward)

- Weekly data review.
- Per the agreed SLA on queries.
- Quarterly backup-restore drill with the technical lead.
- Quarterly review of new validation rules to add (the trial always teaches you something).

When to update the workbook after go-live

Validation rules can be re-imported any time you update the `validation_rules` tab; the audit log captures every change. For changes to forms or team membership, ask the technical lead to make the change directly in the application's admin interface (the importer is one-way for those changes today). The roadmap in the Technical-Lead Guide discusses when bidirectional sync is expected.

Chapter 9. Common mistakes to avoid

Tab names must match exactly. `Study_Users`, not `study_users` or `StudyUsers` or `Users`. Same for the other two tabs.

Field names must match exactly between tabs. If you write `mmse_total` in `Data_Dictionary`, every reference in `validation_rules` must also be `mmse_total` (not `MMSE_total`, not `mmse-total`).

No empty rows in the middle of a tab. The importer reads until it hits the first empty row, then stops. If you delete a row, delete the cells, do not just clear the contents.

Do not put real patient data in the workbook. The workbook is for *configuration* (forms, rules, users), not subject data. Subject data goes into ZZedc directly through the application.

Initial passwords are temporary. Do not use the same password as a real account. The user is forced to change it on first login.

Validation rule expressions need exact spelling. `between 0 and 30` works. `Between 0 and 30` (capital B) might not. `between 0 to 30` definitely will not. Stick to the patterns in §3.6.

Site-id values must be consistent. If you call the site `UCSD` in `Study_Users`, do not call it `Ucsd` in the `Sites` sub-tab. The string match is exact and case-sensitive.

Adding a site after go-live works; renaming one does not. Do not change a `site_id` after data exists for that site. Adding a new site is fine; rename it before any subject is enrolled there, or live with the original name forever.

One coordinator per site is risky. Plan for at least two people per site who can log in, even if only one is the primary data entry person. Vacation, illness, and turnover happen.

Chapter 10. Frequently asked questions

Q. Do I need to use Google Sheets, or can I use Excel?

You can give the technical lead a CSV-shaped folder with `Study_Users.csv`, `Data_Dictionary.csv`, and `validation_rules.csv`. The technical lead has an alternate import path (see the Technical-Lead Guide §"CSV authoring path") that does not require Google. Google Sheets is the easier option because it gives you live editing and collaboration. CSV works for sites that block Google Drive.

Q. Can two of us edit the workbook at the same time?

Yes. Google Sheets handles concurrent editing well. Coordinate who is editing which tab to avoid stepping on each other's work, but the tool itself will not lose your edits.

Q. What happens if I get the validation DSL syntax wrong?

The technical lead will see the parsing error in their import output and tell you which rule failed. Common fixes: change `Between` to `between`, change `to` to `and` in a range expression, and check for typos in field names.

Q. Can I have a field that depends on another field?

Yes. Use the conditional pattern in §3.6:

```
if pregnant == "Yes" then sex == "Female" endif
```

This expresses "if the subject is pregnant, sex must be Female". You can chain conditions and combine them with `and` and `or`.

Q. How do I version-control the workbook?

Google Sheets automatic version history is sufficient for most studies. *File → Version history → See version history* shows every edit. For higher-stakes trials, ask the technical lead to set up a daily snapshot of the workbook into the trial's document repository.

Q. We are migrating from REDCap. Do I still need to do all this?

Less of it. The technical lead can use `import_redcap_to_zzedc_db()` to migrate metadata and audit log. You will still want to walk through the dry-run protocol in Chapter 4 to confirm the migration produced the trial you expected. See the REDCap migration roadmap.

Q. The technical lead says my workbook has 'skipped rules'. What does that mean?

A skipped rule is one the importer did not understand. The technical lead will give you the list. Two things to do: pick each one off and rewrite it using the patterns in §3.6, or flag it for manual implementation by the technical lead. Most skipped rules can be rewritten in five minutes once you see the patterns.

Q. A site refuses to use the application. What now?

Find out why. The most common reasons are: institutional firewall blocks the URL (technical-lead problem); browser too old (replace it); confused by the interface (more training); they liked the previous EDC system (the protocol's choice is final, but the more they understand *why* this system was chosen, the more they will accept it).

Glossary

- **Audit log**: an immutable record of every change made to the database. Every save, edit, query, and export is logged with the user, timestamp, and what changed.
- **CRF** (Case Report Form): a form on which study data are collected.
- **DSL** (Domain-Specific Language): the small plain-English syntax used in the `rule_dsl` column.
- **Field**: a single data-entry box on a form.
- **Form**: a collection of fields filled in together.
- **Lock**: marking a visit as final, blocking further edits without a documented data-correction request.
- **PI** (Principal Investigator): the study lead.
- **Query**: a question raised about a specific data point; resolved by the site coordinator.
- **Role**: the permission level assigned to a user (Admin, PI, Coordinator, Researcher, Monitor, Data-Manager).
- **SAE** (Serious Adverse Event): an adverse event meeting one of six regulatory seriousness criteria.
- **Site**: a physical or institutional location collecting data; identified by a `site_id`.

- **Validation rule**: a check that fires when a user enters data, blocking the save or warning the user if the data fail the check.

Companion guides

- Technical-Lead Guide — the parallel guide for your technical colleague
- Demonstration Trial via Google Sheets — a worked example using a synthetic memory study
- REDCap Migration Roadmap — only relevant if you are migrating from REDCap